MILITARY SPECIFICATION

SAMPLER, AIR PARTICLE, HD-732/UD

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable, line-operated, air particle sampler, hereinafter referred to as the sampler, for the collection of controlled samples of particulate material from the air. The sample, so collected, will be used in conjunction with procedures and equipment not under the scope of specification to determine the extent to which the sampled air was contaminated by radioactive particles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-C-28777/1	Cable Assembly, (Type I), Electronic Test Equipment, (2 poles, 3 Wires, 125 Volts, 13 Amperes, 50/60 Hertz), Grounding Plug Connector And MIL-C-5015 Connector With Cable Clamp
MIL-F-37956	Filter Element, Agent Sampling Extractor

STANDARDS

MILITARY

MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-965	Parts Control Program
DoD-STD-1399 Section 300	Interface Standard For Shipboard Systems Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOK

MIL-HDBK-217	Reliability Prediction Of Electronic Equipment

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS RE-D2696481	Filter Paper Holder
NAVSHIPS RE-D2696482	Protective Cap
NAVSHIPS RE-F2696779	Filter Holder

PUBLICATIONS

ASSISTANT SECRETARY OF THE NAVY (S&L)

NAVMAT P4855-1	Navy Power Supply Reliability And Design And Manufacturing Guidelines

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536E	Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections

(Copies of specifications, standards, handbook, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The sampler shall be capable of drawing a preset volume of air through filter paper for the purpose of collecting airborne particulate. The sampler shall consist of a blower (enclosed in a housing), flow totalizer and accessories, including two filter paper holders, two filter holders, and an inlet nozzle protective cap. The sampler shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Major component list. Major components shall be as specified in 3.1.2.1 through 3.1.2.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

3.1.2.1 Flow totalizer. The flow totalizer shall be located at the inlet and shall measure and indicate the volume of air being drawn through the filter paper by the blower. The indication shall be in cubic meters (m^3) and tenths of m^3 up to a maximum of 99.9 m^3. Upon reaching this maximum, the totalizer shall continue to indicate starting again from zero. The indicating portion of the totalizer shall be a register capable of being reset from any point within the range of indication. A means of adjustment shall be provided for calibration of the flow totalizer. This adjustment shall be accessible from the exterior of the flow totalizer. The blower shall shut off when the preset volume of air has been drawn through the filter paper.

3.1.2.2 Housing. The housing shall be constructed of steel or cast aluminum and or both shall include a power switch, fuses, and all controls and indicators, and electronics. The power and control switches and fuses shall be mounted on the exterior of the housing in a recessed or enclosed panel to prevent damage or inadvertent movement. A handle shall be provided and positioned such that the equipment balances easily in the hand. The housing shall also be provided with antiskid rubber feet. The design of the housing shall be approved by the procuring activity. There shall be an outlet nozzle for the exhaust that shall have an inner diameter of 0.8125 inches (in.) $\pm$0.0312 in. and an outer diameter of 1.0 in. $\pm$0.0312 in. The length of the outlet nozzle shall be between 1.5 in. and 2 in.

3.1.2.3 Filter paper holder. The filter paper holder shall be in accordance with NAVSHIPS Drawing (DWG) RE-D2696481. Two filter paper holders shall be provided with each sampler.

3.1.2.4 Filter holder. The filter holder shall be compatible with the flow totalizer and shall be in accordance with NAVSHIPS DWG RE-F2696779. Two filter holders shall be provided with each sampler.

3.1.2.5 Protective cap. The protective cap shall be compatible with the flow totalizer and shall be in accordance with RE-D2696482. The protective cap shall cover the inlet nozzle when the filter holder and filter paper holder are removed. One protective cap shall be provided with each sampler.

3.1.2.6 Filter paper. The sampler is intended for use with filter paper conforming to the requirements of MIL-F-37956. The sampler shall conform to the requirements of this specification when used with this filter paper. The design shall be based on use of the paper specified herein. All performance tests, to determine compliance with this specification, shall be made with this filter paper. Filter paper will not be required as part of the equipment covered by this specification.

3.1.2.7 Motor and blower. The motor and blower shall conform to the performance characteristics specified in 3.2.1. The motor shall conform to the Motors, dynamotors, rotary power converters, and motor-generators paragraph of MIL-E-16400.

3.1.2.8 Stowage case. A rectangular stowage case for the sampler shall be provided. The stowage case shall be constructed of painted steel and shall closely fit the sampler, being not greater than 15 in. in height, 15 in. in

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

length, and 15 in. in depth. The stowage case shall include fittings which securely fit the sampler (with a filter paper holder in place) in position with the case lid closed. The stowage case shall include a padded canvas (or rugged fabric) bag containing one spare filter holder and a package for containing not less than 50 replacement filter papers and 50 filter paper envelopes. The bag shall have a rugged drawstring of sufficient length to permit easy handling by the operator and to permit the bag to be tied to the sampler handle. The stowage case shall have a top lid hinged along one dimension with a full length piano hinge. Opposite the hinge, a hasp and a captured hasp locking bar shall be provided. The stowage case shall also include tie-down rings or bars on two opposite sides to permit the stowage case to be secured with nylon straps to a deck without interfering with the lid.

3.1.3 Government furnished property. The Government will provide the quantities of items specified in a through d:

a. 150 filter papers
b. Two filter paper holders
c. Two filter holders
d. One protective cap

3.2 Characteristics. Characteristics for the sampler shall be as specified in 3.2.1 through 3.2.4.1.

3.2.1 Performance characteristics. The performance characteristics shall be as specified in 3.2.1.1 through 3.2.1.15

3.2.1.1 Flow rate. The flow rate, with a filter paper installed, shall be between 0.3 cubic meters per minute (m^3/min) and 0.5 m^3/min inclusive.

3.2.1.2 Air flow performance. At 115 volts (V), 60 hertz (Hz) the sampler shall draw at least 0.3 m^3/min with a static pressure of 33 millimeters (mm) of mercury (Hg). At 103 V, 57 Hz the sampler shall draw at least 0.2 m^3/min with a static pressure of 33 mm Hg.

3.2.1.3 Totalizer accuracy. The accuracy of the flow totalizer shall be within ±10 percent of the actual volume of air drawn through the filter paper. A means of adjustment shall be provided for calibration of the flow totalizer. The calibration control for the flow totalizer shall be internal to the housing of the sampler.

3.2.1.4 Controls and indicators. Controls and indicators shall include a power switch, a totalizer indicator, a presettable air volume switch, a start button, a stop button, and an override switch for the presettable start/stop function.

3.2.1.5 Temperature classification. The sampler shall conform to the operating and nonoperating requirements without adjustment under Range 2 temperature conditions of MIL-E-16400.

3.2.1.6 Humidity. The sampler shall conform to the Humidity paragraph of MIL-E-16400.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.

3.2.1.7 Shock. The sampler shall be subjected to Condition I of Test Method 213B, shock, medium impact test of MIL-STD-202, and shall conform to the requirements of MIL-S-901 for Grade B, Type A, Class I.

3.2.1.8 Inclination. The sampler shall conform to the Inclination paragraph of MIL-E-16400 for submarine installation.

3.2.1.9 Airborne noise. The airborne noise radiated from the sampler shall not exceed Grade D of MIL-STD-740.

3.2.1.10 Enclosure. The enclosure shall be in accordance with MIL-STD-108. Degree of enclosure shall be dripproof (45 degrees).

3.2.1.11 Electrical power characteristics. Electrical power characteristics shall be as specified in 3.2.1.11.1 and 3.2.1.11.2.

3.2.1.11.1 Shipboard alternating current (AC) power source. The sampler shall conform to the Shipboard a.c. power source paragraph of MIL-E-16400 and 3.2.1.2, operating from a voltage supply of 115 V ±10 percent, 60 Hz, ±5 percent, single-phase, ungrounded, Type I shipboard power as specified in DoD-STD-1399 Section 300 and shall conform to the user equipment requirements specified in this standard. The running current shall not exceed 8.0 amperes.

3.2.1.11.2 Fixed shore station AC power source. The equipment shall conform to the Fixed shore station a.c. power source paragraph of MIL-E-16400 and 3.2.1.2, operating from a voltage supply of 115 V ±10 percent, 60 Hz ±5 percent, single phase. The running current shall not exceed 8.0 amperes.

3.2.1.12 Leakage current (equipment). The leakage current shall conform to the Leakage current (equipment) paragraph of MIL-E-16400. The leakage current shall not exceed 5 milliamperes (mA).

3.2.1.13 Dielectric withstanding voltage. The dielectric withstanding voltage shall conform to the Dielectric withstanding voltage paragraph of MIL-E-16400.

3.2.1.14 Interchangeability. Individual components specified in 3.1.2 shall be interchangeable in accordance with of MIL-STD-454, Requirement 7. No adjustment or calibration shall be allowed after an interchange of the individual components specified in 3.1.2.2, 3.1.2.3, 3.1.2.4, 3.1.2.5, 3.1.2.6, and 3.1.2.8 has been made.

3.2.1.15 Vibration. The sampler shall conform to the requirements of MIL-STD-167-1 for Type I environmental vibration.

3.2.2 Physical characteristics. Physical characteristics shall be as specified in 3.2.2.1 through 3.2.2.4.

3.2.2.1 Weight. The weight of the sampler shall not exceed 18 pounds (lbs). The weight of the sampler, accessories, and stowage case shall not exceed 30 lbs.

3.2.2.2 Dimensions and volume. The dimensions of the sampler shall not exceed 13 in. in height (exclusive of the carrying handle), 13 in. in length (with a

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

filter holder in place), and 13 in. in depth.

3.2.2.3 Parts, materials, and processes. Parts, materials, and processes shall conform to the Parts, materials, and processes paragraph of MIL-E-16400. Parts control shall be in accordance with MIL-STD-965, Procedure I.

3.2.2.4 Identification and marking. Identification and marking shall conform to the Identification and marking paragraph of MIL-E-16400.

3.2.3 Reliability. Reliability shall be as specified in 3.2.3.1.

3.2.3.1 Mean-time-between-failures (MTBF). The upper test MTBF (θ_0 as defined by MIL-STD-781) shall be 1000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 500 hours.

3.2.4 Maintainability. Maintainability shall be as specified in 3.2.4.1.

3.2.4.1 Quantitative corrective maintenance. The sampler shall have a mean-corrective-maintenance-time not exceeding 0.5 hours and a maximum-corrective-maintenance-time (95th percentile) not exceeding 1.0 hours.

3.3 Design and construction. Design and construction shall be as specified in 3.3.1 through 3.3.6.

3.3.1 Electromagnetic environmental effects (E^3). The sampler shall be self-compatible and shall operate compatibly in an electromagnetic environment. The sampler shall conform to, as a minimum, the requirements of MIL-STD-461 for Class A4 equipment. Requirements CE01, CE03, CS06, RE01 (the limits of which are modified as shown in FIGURE 1, RE02, RS01, and RS02 (Part II only) apply.

3.3.2 Workmanship. Workmanship shall be as specified in 3.3.2.1 through 3.3.2.3.

3.3.2.1 General workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9.

3.3.2.2 Workmanship screen. All samplers shall withstand a defect detection vibration screen (see 4.5.3.1) of random type vibration at 0.04 g^2/Hz $\pm$3 decibels (dB) from 80 Hz to 350 Hz and temperature cycling (see 4.5.3.2) with operation at the maximum and minimum operating temperatures and dwell time at the nonoperating temperature extremes.

3.3.2.3 Soldering and soldering processes. Soldering, soldering procedures and processes shall conform to WS-6536E.

3.3.3 Safety. Safety shall be as specified in 3.3.3.1 and 3.3.3.2.

3.3.3.1 Circuit breakers, fuses, and fuse holders. Circuit breakers, fuses, and fuse holders shall conform to the Circuit breakers and the Fuses and fuse holders paragraphs of MIL-E-16400. Both sides of the power input shall be protected. Circuit breakers or fuses shall be capable of protecting the motor under locked rotor conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.

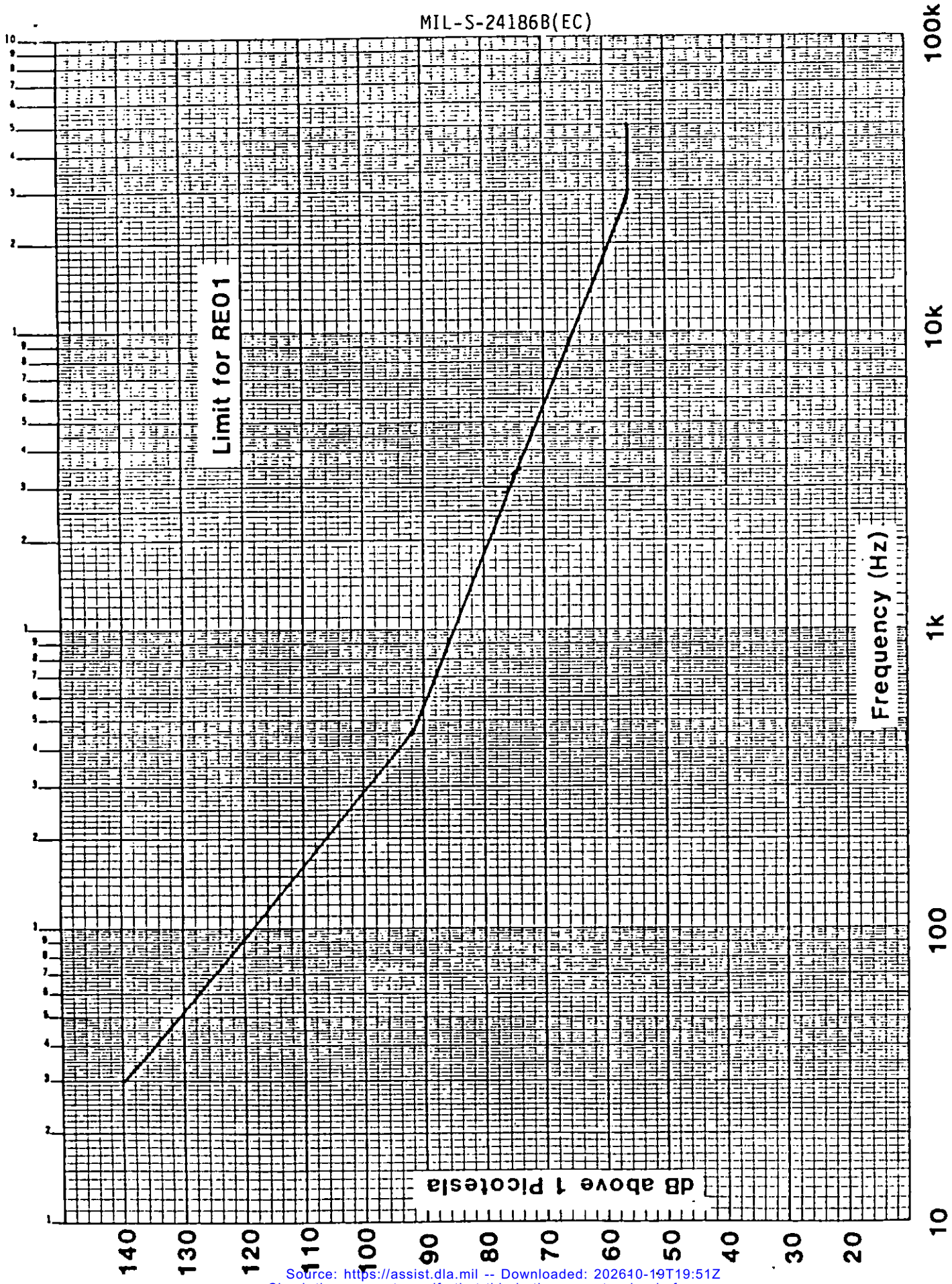


FIGURE 1. Limits for RE01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

3.3.3.2 Power cable. The power cable shall conform to the requirements of MIL-C-28777/1. The cable length shall be 15 ±0.25 feet (ft). A means shall be provided to secure the power cable to the sampler when not in use.

3.3.4 Human engineering. Human engineering shall conform to the human engineering operational and maintenance design criteria of MIL-STD-1472.

3.3.5 Power supplies. Design for low voltage power supplies delivering up to 5 kilowatts at 300 volts direct current, or less shall conform to the following requirements specified in 3.3.5.1 through 3.3.5.9. Additional guidance for power supplies is specified in NAVMAT P4855-1.

3.3.5.1 Power density. The power density is the output power divided by power supply envelope volume, including electromagnetic interference (EMI) filtering, where required. Power density shall not exceed 2 watts per cubic inch.

3.3.5.2 Junction temperature. Junction temperature of semiconductor and microelectronic devices (diodes, transistors, hybrid and integrated circuits, and so forth) shall not exceed +110° Celsius (C) under worst-case conditions.

3.3.5.3 Case/hot spot temperatures. Case/hot spot temperatures shall not exceed 40°C rise above ambient temperature with a maximum temperature of +110°C for parts less than or equal to 3 watts of dissipation; 55°C rise above ambient with a maximum temperature of +125°C for parts greater than 3 watts of dissipation; 30°C for transformers; and 10°C rise above ambient, due to self-heating, with a maximum temperature of 85°C for capacitors.

3.3.5.4 Maintainability. Power supplies shall be designed to be maintainable at either intermediate or depot level; the power supplies shall not be limited by design to repair by manufacturer.

3.3.5.5 Encapsulation. Power supplies shall not be encapsulated or embedded (potted) unless it can be shown by analyses and test to be necessary for heat removal or dissipation. This requirement shall not exclude conformal coating.

3.3.5.6 Workmanship. Power supply manufacturing shall include random vibration and temperature cycling of every unit, under full electrical load; this may be done in the end item but shall be performed on a unit basis for spares.

3.3.5.7 Power. Samplers incorporating power supplies shall be designed to operate from Type I power specified in DoD-STD-1399, Section 300. Power supplies shall be tested with the end equipment, or shall be tested as a unit by duplicating the input and output power of the power supply as installed for operation in the end equipment.

3.3.5.8 Power supply reliability. The MTBF of the power supply shall be not less than 40,000 hours at 55°C.

3.3.5.9 Load. The power supply shall be designed to withstand and shall not be damaged by any load between an open circuit and a short circuit. The power supply shall be designed with a reserve load capacity of 20 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

3.3.6 Transportability. The sampler shall be designed to withstand the environmental problems resulting from normal transportation and handling without damage or calibration change. Calibration and alignment changes that can be corrected using only the front operators controls will be acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
    1. Production inspection (see 4.4.1)
c. Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. Unless otherwise specified (see 6.2) four units shall be required for first article inspection. First article inspection shall

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.

consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Surface examination	3.1.2	4.5.1
Weight	3.2.2.1	4.5.1
Dimensions and volume	3.2.2.2	4.5.1
Parts, materials and processes	3.2.2.3	4.5.1
Identification and marking	3.2.2.4	4.5.1
Safety	3.3.3	4.5.1
General workmanship	3.3.2.1	4.5.2
Workmanship screen	3.3.2.2	4.5.3
Soldering and soldering processes	3.3.2.3	4.5.2
Performance		
Flow rate	3.2.1.1	4.5.4
Air flow performance	3.2.1.2	4.5.5
Totalizer accuracy	3.2.1.3	4.5.6
Controls and indicators	3.2.1.4	4.5.1
Temperature	3.2.1.5	4.5.1
Humidity	3.2.1.6	4.5.1
Shock	3.2.1.7	4.5.7
Inclination	3.2.1.8	4.5.1
Airborne noise	3.2.1.9	4.5.1
Enclosure	3.2.1.10	4.5.8
Electrical power characteristics		
Power and power factor	3.2.1.11.1	4.5.1
Steady state voltage and frequency	3.2.1.11.1	4.5.1
Transient voltage and frequency	3.2.1.11.1	4.5.1
Spike voltage and power interruption	3.2.1.11.1	4.5.1
Harmonic current	3.3.1.11.1	4.5.1
Leakage current	3.2.1.12	4.5.9
Dielectric withstanding voltage	3.2.1.13	4.5.1
Interchangeability	3.2.1.14	4.5.10
Vibration	3.2.1.15	4.5.16
Reliability	3.2.3	4.5.12
Maintainability	3.2.4	4.5.13
E[3]	3.3.1	4.5.11
Human engineering	3.3.4	4.5.15

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection. (continued)

Examination or test	Requirement paragraph	Test paragraph
Power supply		
Power density	3.3.5.1	4.5.14.1
Junction temperature	3.3.5.2	4.5.14.2
Case/hot spot temperature	3.3.5.3	4.5.14.3
Maintainability	3.3.5.4	4.5.13
Encapsulation	3.3.5.5	4.5.14.4
Workmanship	3.3.5.6	4.5.14.5
Power	3.3.5.7	4.5.14.6
Power supply reliability	3.3.5.8	4.5.14.7
Load	3.3.5.9	4.5.14.8

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1.

4.4.1 Production inspection. Production inspection shall comprise examinations and tests which will prove and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect defects of materials. Production inspection shall include the examinations and tests specified in TABLE II.

TABLE II. Production inspection.

Examination or test	Requirement paragraph	Test paragraph	Sample size
Surface examination	3.1.2	4.5.1	All
Weight	3.2.2.1	4.5.1	All
Dimensions and volume	3.2.2.2	4.5.1	All
Parts, materials and processes	3.2.2.3	4.5.1	All
Identification and marking	3.2.2.4	4.5.1	All
Safety	3.3.3	4.5.1	All
General workmanship	3.3.2.1	4.5.2	All
Workmanship screen	3.3.2.2	4.5.3	All
Performance			
Flow rate	3.2.1.1	4.5.4	All
Air flow performance	3.2.1.2	4.5.5	All
Totalizer accuracy	3.2.1.3	4.5.6	All
Leakage current	3.2.1.12	4.5.9	2 per lot
E[3]	3.3.1	4.5.11	2 per lot

Source: https://assist.dla.mil -- Downloaded: 202640-19T19:51Z
Check the source to verify that this is the current version before use.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.16.

4.5.1 Examinations and tests. Examinations and tests specified in TABLE I and TABLE II shall be conducted in accordance with the applicable test of MIL-E-16400, except as specified in 4.5.2 through 4.5.16.

4.5.2 General workmanship. The sampler, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of WS-6536E. Each solder connection and its associated wiring or leads shall be visually examined.

4.5.3 Workmanship screen. Vibration and temperature cycling shall be performed on each sampler as specified in 4.5.3.1 and 4.5.3.2. Vibration shall be performed prior to temperature cycling.

4.5.3.1 Vibration. Vibration shall be as specified in a through d.

a. The vibration test may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration.

b. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 2.

c. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for the vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the root-mean-square (rms) acceleration level.

d. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

4.5.3.2 Temperature cycling. Each sampler shall be subjected to 10 cycles of the temperature curve in shown FIGURE 3. The temperature rate of change shall be not less that 5°C per minute. Sampler power shall be turned on and off at the indicated times. The sampler shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the sampler reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

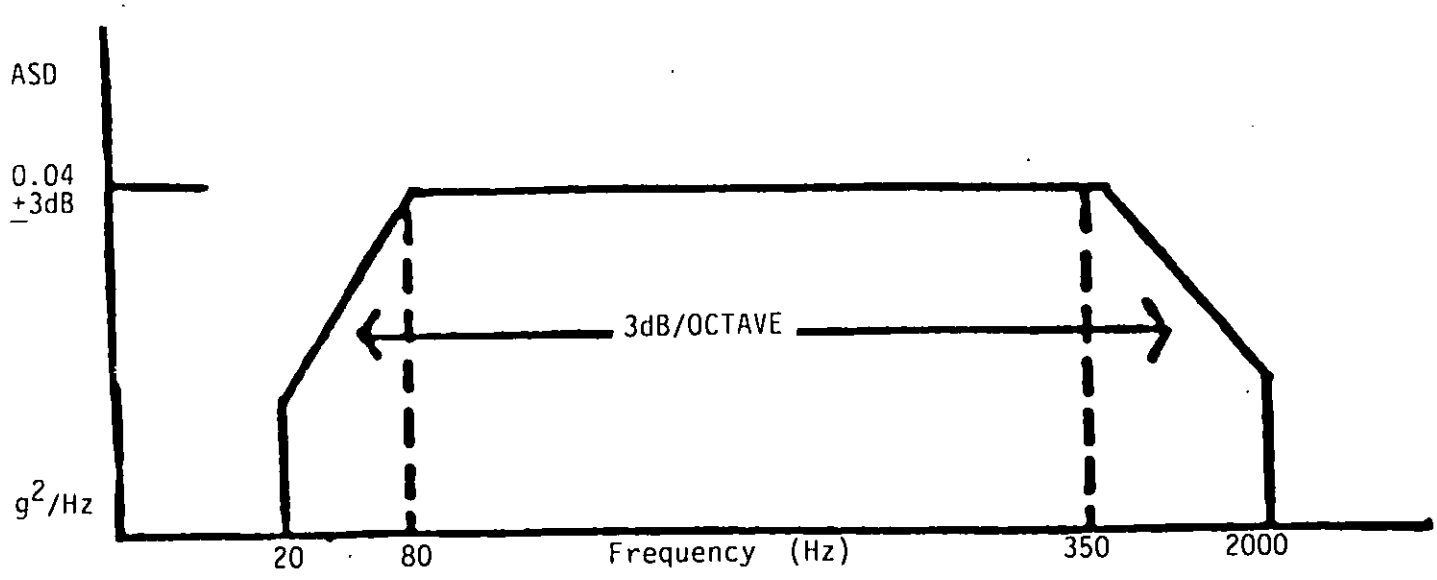


FIGURE 2. Random vibration curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

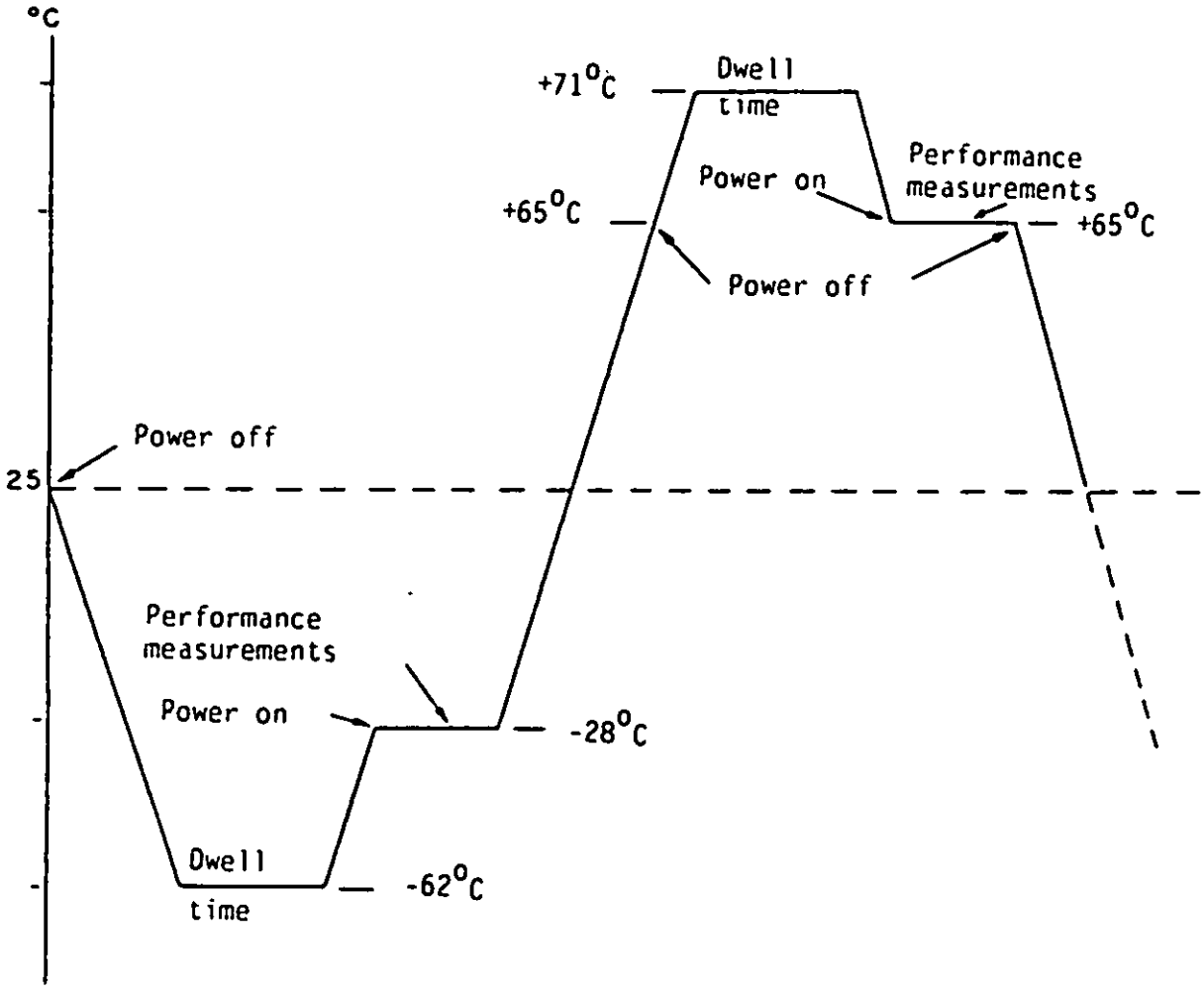


FIGURE 3. One cycle of temperature curve (not to scale)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

4.5.4 Flow rate. A calibrated National Bureau of Standards (NBS) flowmeter traceable to ±5 percent at the 95 percent confidence level shall be used. The flowmeter shall be coupled to the sampler. The measurement of the flow rate shall be conducted with a filter paper in place. The reading of the sampler shall conform to the requirements of 3.2.1.1 (after setting of calibration adjustment).

4.5.5 Air flow performance. Air flow performance tests shall be conducted using a calibrated NBS pressure gage traceable to ±5 percent at the 95 percent confidence level and a calibrated flowmeter (NBS traceable to ±5 percent at the 95 percent confidence level). The results shall conform to the requirements of 3.2.1.2.

4.5.6 Totalizer accuracy. The calibration test shall be conducted using the method of 4.5.4. The totalizer reading shall be within ±10 percent of the reading on the calibration standard. The sampling times shall be 15 minutes and 30 minutes.

4.5.7 Shock. The sampler shall be tested to verify conformance to the requirements of 3.2.1.7. For the test of MIL-STD-202, the sampler shall be dropped on only one axis (with the bottom of the unit facing downwards) three consecutive times. There shall be no more than 5 percent degradation in performance and the sampler shall remain operable without mechanical deterioration. The shock test shall be conducted once on the sampler out of its stowage case and conducted once with the sampler inside its stowage case. Outside its stowage case, the sampler shall be dropped from a height of 40 in. The sampler inside its stowage case shall be dropped from a height of 5 ft. For the test of MIL-S-901, the test shall be conducted as a Grade B, Type A, Class 1. The test shall be conducted with the sampler in its stowage case.

4.5.8. Enclosure. The sampler shall be tested to verify conformance to the requirements of 3.2.1.10. The sampler shall not be tilted in a backward position (with the air intake facing upward).

4.5.9 Leakage current. The leakage current shall be measured at 115 V., 60 Hz, single-phase. During first article inspection (see 4.3), this test shall be performed after the vibration test (see 4.5.16) and shock test of MIL-STD-202 (see 4.5.7).

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE SAMPLER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.9.1 Sampler test connection. After power removal, each sampler directly connected to an external power source and units deriving power from the sampler shall be placed on an insulated surface. All safety ground conductors between the sampler shall be intact. The safety ground conductor between the sampler and the source power shall be opened during the test. OBSERVE WARNING STATEMENT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

The sampler shall be connected as shown in FIGURE 4 for single-phase source power.

4.5.9.2 Measurement. Leakage current shall be measured on sampler in its normal operating configuration. Sampler controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the sampler is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.10 Interchangeability demonstration. The sampler shall be examined and tested to demonstrate compliance with 3.2.1.14 by utilizing at least two samplers and interchanging each interchangeable assembly, subassembly, module, and so forth, between the two samplers. Satisfactory sampler performance shall be demonstrated prior to and after the interchange of the interchangeable assemblies, subassemblies, and so forth. Interchangeability of piece parts shall be verified by examination.

4.5.11 E^3. Conformance to the requirements of 3.3.1 shall be verified by the test procedures of MIL-STD-462.

4.5.12 Reliability. When required by the procuring activity (see 6.2), a production reliability acceptance test (PRAT) shall be conducted in accordance with the requirements of MIL-STD-781 except as specified in a through c:

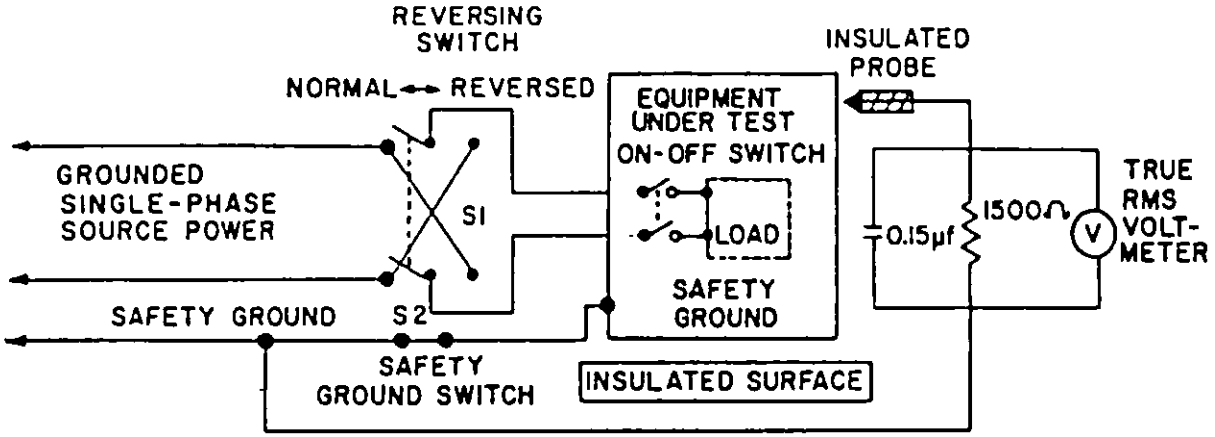
a. The PRAT shall proceed in accordance with the accept or reject criteria of MIL-STD-781, Test Plan VIIIC. Only ON time shall be counted.
b. The PRAT environmental conditions shall be in accordance with MIL-STD-781 equipment Category 3A and the Test condition matrix Table for shipboard sheltered equipments.
c. The sampler ON-OFF cycling profile shall be 3 minutes OFF (de-energized) and 3 minutes ON (energized), rendering a duty cycle of 50 percent.

4.5.13 Maintainability. Maintainability testing shall be in accordance with MIL-STD-471.

4.5.14 Power supply. The test methods specified in 4.5.14.1 through 4.5.14.8 apply to the power supply.

4.5.14.1 Power density. The power density shall be tested by dividing the output power by the power supply envelope volume, including EMI filtering where required. The power density shall not exceed 2 watts per cubic meter.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.



GENERAL ORDER OF TEST:

1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. S1 SW normal. S2 SW closed. Connect source power.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. S1 SW REVERSED. ON-OFF SW ON. Repeat Step 4.
7. ON-OFF SW OFF. Repeat Step 4.
8. S2 SW CLOSED. S1 SW normal.
9. Repeat Step 3 THRU 8 for each mode of operation.
10. Remove source power. Disconnect equipment. Sign record sheet.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Single-phase test diagram for leakage current measurement.

Source: http://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

4.5.14.2 Junction temperature. Testing for junction temperatures shall be accomplished in accordance with MIL-HDBK-217 and the Thermal test paragraph of MIL-E-16400.

4.5.14.3 Case/hot spot temperatures. Testing for case/hot spot temperatures shall be accomplished in accordance with MIL-HDBK-217 and the Thermal test paragraph of MIL-E-16400.

4.5.14.4 Encapsulation. Encapsulation of the power supplies shall be tested through visual examination.

4.5.14.5 Workmanship. The testing for power supply manufacturing shall be tested in accordance with 4.5.3.

4.5.14.6 Power. Power requirements for the power supply shall be tested in accordance with DoD-STD-1399, Section 300.

4.5.14.7 Power supply reliability. As required by the procuring activity (see 6.2), an analysis shall be conducted as specified (see 6.2) to verify conformance to requirements of 3.3.5.8.

4.3.14.8 Load. Load requirements shall be tested by:

a. Making no connections between the output of the power supply, leaving the power supply running for a minimum of 30 minutes, and then verifying that the power supply runs satisfactorily under normal load conditions for the sampler for a minimum of 30 minutes

b. Shorting the output of the power supply with a load of 1-ohm (maximum), running the power supply for a minimum of 30 minutes, and then verifying that the power supply runs satisfactorily under normal sampler load conditions for a minimum of 30 minutes

c. Shorting the output of the power supply using a suitable impedance to cause the power supply to deliver 20 percent more power than is normally required for the sampler and running the power supply for a minimum of 30 minutes. Then, proper operation of the power supply shall be verified by running the power supply under normal conditions for a minimum of 30 minutes.

4.5.15 Human Engineering. The sampler shall be tested to verify conformance to the requirements of 3.3.4.

4.5.16 Vibration. The sampler shall be tested to verify conformance to the requirements of 3.2.1.15.

4.6 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the

Source: https://assist.dla.mil -- Downloaded: 202640-19T19:51Z
Check the source to verify that this is the currentversion before use.

contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The sampler, covered by this specification, is intended for use on board ships and in shore facilities when continuous monitoring of airborne particulate radioactivity is required.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. If the reliability test is to be conducted in accordance with MIL-STD-781 (see 4.5.12)
d. Method of analysis of power supply reliability (see 4.5.14.7)
e. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the items should be first article samples. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.1.3.

6.5 Subject term (keyword) listing.

Distribution statement A
Electromagnetic interference (EMI)
Encapsulation
HD-732/UD
Human engineering
Intended use (shipboard and shore)
Interchangeability
Leakage current
Maintainability
Random vibration
Reliability

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:51Z
Check the source to verify that this is the current version before use.

Safety
Sampler, air particle
Shock
Soldering
Temperature cycling
Transportability
Vibration
Workmanship screen

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC
(Project no. 6625-N866(EC))

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

SPACE AND NAVAL WARFARE SYSTEMS COMMAND
WASHINGTON, D.C. 20363-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
SPACE AND NAVAL WARFARE SYSTEMS COMMAND
ATTN: SPAWAR 003-121
WASHINGTON, DC 20363-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24186B(EC)	SAMPLER, AIR PARTICLE, HD-732/UD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-01-19 19:51Z
Check the source to verify that this is the current version before use.